

Why Python is a Dynamically Typed Language

Python is called a **dynamically typed language** because the type of a variable is determined at **runtime** rather than when the code is written. A variable can hold values of different data types during program execution without requiring explicit type declarations.

Example:

```
x = 10          # x is an integer
print(type(x))

x = "Hello"     # x is now a string
print(type(x))
```

Output:

```
<class 'int'>
<class 'str'>
```

In the above example, the same variable `x` first stores an integer and later stores a string. Python automatically identifies the data type during execution.

Advantages of Dynamic Typing

1. **Easy to write and read code**
2. **Less code required** since data types need not be declared
3. **Faster development and prototyping**
4. **Greater flexibility** in handling different types of data

Disadvantages of Dynamic Typing

1. Type-related errors are detected only at runtime.
2. Can make debugging more difficult in large applications.
3. Generally slower than statically typed languages due to runtime type checking.